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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2022
Course Title: Inorganic Chemistry III (Advanced Concepts)
Course No.: Chem-320F
Full Marks: 70, Time: 4 hours

Answer any five (5) questions

1. (a) What is a *wave function*? Deduce the general expression for *Schrödinger wave equation* for a standing wave motion of a particle with mass *m* defining each term in it. 1 + 2 = 3

(b) What are Radial and Angular wave functions? Provide a graphical comparison between Radial wave function (R) and Radial Probability function (r^2R^2) for 1*s* orbital of hydrogen. 2 + 2 = 4

(c) Why the wave function Ψ is squared? Point out the significance of Ψ and Ψ^2. 2

(d) (i) Define Angular function $\theta(\theta)$ $\phi(\phi)$ and Radial function $R(r)$. What part of an orbital is signified by these two factors? 2.5

(ii) Show that if $\Psi = A \sin rx + B \cos sx$ (where A, B, r and s are constants), then the wave equation for a particle in a 1-dimensional box is $\Psi = A \sin rx$. 2.5

2. (a) Explain the formation of the followings on the basis of VBT: 4.5
(i) paramagnetic $[FeF_6]^{3-}$, (ii) diamagnetic $[Ni(CN)_4]^{2-}$, (iii) tetrahedral $[NiCl_4]^{2-}$

(b) What are inner and outer complexes? Inner complexes are stable while the outer complexes are labile – explain why. 2.5

(c) Discuss the origin of magnetic property in coordination complexes. 3

(d) Calculate the spin only magnetic moment for Mn^{2+} and Fe^{3+} complexes. 2

(e) Why the magnetic moment of an octahedral monothiocarbamate complex of iron(III) dropped from 5.8 μB at 300 K to 4 μB at 78 K? 2

3. (a) What are Bonding and Anti-bonding Molecular Orbitals? Write down the bonding and anti-bonding wave functions for the two molecular orbitals. What is overlap integral. 1.5 + 1.5 + 1 = 4

(b) How would you account for the fact that N_2 is unreactive and diamagnetic but molecular O_2 is paramagnetic? 2

(c) Provide a molecular orbital diagram for heteronuclear molecule CO and compare C–O bond order and bond length in C^+ and CO^+. 2

(d) Define σ and π overlap. Schematically show the π overlap of a metal *d* orbital with the followings: 4

(i) a ligand *p* orbital, (ii) a metal *d* orbital,
(iii) a ligand π^* orbital, (iv) a ligand σ^* orbital

(e) What are Ligand Group Orbitals (LGO)? 2

4. (a) What are the unique properties of transition metal elements? 1
(b) Write down the nomenclature of the following coordination complexes: 3
(i) $[Pt(NH_3)BrCl(CH_3NH_2)]$, (ii) $[Co(H_2NCH_2CH_2NH_2)_2Cl_2]^+$,
(iii) $[Co(NH_3)_4(OH)_2]^{3+}$
(c) Define denticity. When the eta (η) or kappa (κ) notations are used in nomenclature? Draw the structure of the following ligands and mention their denticity: 2 + 3 = 5
(i) Trien, (ii) EDTA, (iii) acac, (iv) dppe
(d) Define isomerism. Draw structures of the *fac* and *mer* isomers of the complex $[Co(NH_3)_3Cl_3]$. Write notes on the following with examples: 2 + 3 = 5
(i) Linkage isomer, (ii) Ionization isomer, (iii) Hydrate isomer

5. (a) Draw the shapes of the various *d* orbitals and explain why they split into two groups – t_{2g} and e_g in an octahedral ligand field? Briefly discuss the factors that affect the crystal field splitting energy (Δ). 5
(b) Which complexes of the following pairs has a large Δ value? Explain your answer. 4.5
(i) $[Cr(NH_3)_6]^{3+}$ and $[CrCl_6]^{3-}$
(ii) $[Co(H_2O)_6]^{3+}$ and $[Rh(H_2O)_6]^{3+}$
(iii) $[Cr(H_2O)_6]^{3+}$ and $[Cr(H_2O)_6]^{2+}$
(c) Determine the number of unpaired electrons and CFSE (ignore the pairing energy) of the followings: $[CoCl_4]^{2-}$, $[Zn(CN)_6]^{4-}$, $[Fe(H_2O)_6]^{3+}$ 4.5

6. (a) What is Jahn-Teller theorem? On the basis of Jahn-Teller theorem explain the distortion of six coordinated complexes. 6
(b) Thermodynamic properties of transition metal complexes are affected by *d*-orbital splitting – explain. 5
(c) In the crystal structure of CuF_6, the Cu^{2+} is coordinated by six F^- with four F^- at a distance of 1.93 Å and two F^- at a distance of 2.27 Å – explain why. 3

7. (a) Explain the terms Lability and Inertness of complexes? 4
(b) Show the substitution reaction mechanism in $[Co(NH_3)_5(CO_3)]^+$ to produce $[Co(NH_3)_5(H_2O)]^{3+}$ without breaking metal-ligand bond. 3
(c) Discuss the mechanism of nucleophilic substitution reaction in square planer complexes. 3
(d) What is trans-effect? Explain it in terms of electrostatic polarization theory. 4

8. (a) Deduce the equation $P = 1 + 4\pi\kappa$ and mention the terms in it. What are gram (χ_g) and molar (χ_m) susceptibilities? 2 + 2 = 4
(b) What are anti-ferromagnetic and ferromagnetic coupling? State Curies Law and show change of χT graphically for the above two types of interactions with variable temperature. 3 + 2 = 5
(c) An iron(III) chloride sample has gram susceptibility (χ_g) of 1.28×10^{-2} erg. G^{-2} mol^{-1} at 19.9 °C. Calculate μ_{eff} for the sample and compare with spin only value μ_{so}. 5

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) (Part III) Final Examination 2021
Course Title: Inorganic Chemistry III (Advanced Concepts)
Course No. Chem. 320F
Time: 4 Hours, Total Marks: 70

Answer any five questions

1. (a) What is a wave function? Deduce the general expression for Schrödinger wave equation for a standing wave motion of a particle with mass m defining each term in it. 1+2 = 3
   (b) What are radial and angular wave functions? Provide a graphical comparison between Radial wave function (R) and Radial probability function (r^2R^2) for 1s orbital of Hydrogen. 2+2 = 4
   (c) Why the wave function ψ is squared? Point out the significance of ψ and ψ^2. 2
   (d) (i) Define Angular function θ(θ) ϕ(ϕ) and Radial function R(r). What part of an orbital is signified by these two factors? 2.5
   (ii) Show that if ψ = A sin rx + B cos sx (where A, B, r and s are constants) is the wave equation for a particle in a 1-dimensional box, 2.5

$$\psi = A \sin rx$$

2. (a) What are Bonding and Anti-bonding Molecular orbitals? Write down the bonding and anti-bonding wave functions for the two MOs. What is overlap integral? Draw the (a) probability function diagram for bonding Ψ_b^2 and anti-bonding Ψ_a^2. 1.5+1.5+1 = 4
   (b) How would you account for the fact that, N_2 is unreactive and diamagnetic but molecular O_2 is paramagnetic? 2
   (c) Provide a MO diagram for heterodiatomic molecule CO and compare bond order and bond length in CO and CO^+. 2
   (d) (i) Define π and σ overlap. 4
   (ii) Draw the π-overlap of a metal d orbital with the following
   (A) A ligand p orbital (B) A metal d orbital. (C) ligand π* orbital (D) ligand σ* orbital 2
   (e) What are Ligand group orbital (LGO)?

3. (a) Which complexes of the following pairs have a larger Δ_o value? Explain your answer. 4
   (i) $[Fe(H_2O)_6]^{2+}$ and $[Fe(CN)_6]^{4-}$
   (ii) $[Co(NH_3)_6]^{3+}$ and $[Rh(H_2O)_6]^{3+}$
   (b) Draw the possible isomeric structures of $[Co(NH_3)_3Cl_3]$ and $[Co(dien)_2]^{3+}$. 3
   (c) Write down the names of the following complex or ions 0.5×4 = 2
   (i) $Mo[(C_6H_5)_2PCH_2CH_2P(C_6H_5)_2]_2(N_2)_2$
   (ii) $[Co(Co(NH_3)_4(OH)_2)_3]^{6+}$
   (iii) *mer*-$[Ru\{(C_2H_5)_3P\}_3H_3]$
   (iv) $[(NH_3)_5CoNH_2Co(NH_3)_4(H_2O)]Cl_5$
   (d) What is crystal field stabilization energy? Explain the splitting of *d*-orbitals under the influence of an octahedral field. 3
   (e) Find out the ~~exchange~~ CFSE energy for a d^7 ion, both as a high-spin and as a low-spin in an octahedral complex. 2

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4. (a) What is Jahn-Teller theorem? On the basis of Jahn-Teller theorem explain the distortion of six coordinated complexes.
   (b) Thermodynamic properties of transition metal complexes are affected by d-orbital splitting. - Explain.
   (c) Ethylenediamine (en) forms stable chelate with Cu^{2+} ion, yet $[Cu(en)_3]^{2+}$ ion is highly unstable. – Explain.

5. (a) Complexes of monovalent silver are diamagnetic, of bivalent silver are paramagnetic and of trivalent silver again diamagnetic. Explain.
   (b) The high-spin d^4 complex $[Cr(H_2O)_6]^{2+}$ is *labile*, but the low-spin d^4 complex ion $[Cr(CN)_6]^{4-}$ is *inert*. Clarify this statement. 3
   (c) Using the angular overlap model, determine the relative energies of *d* orbitals in a metal complex of formula ML_4 having tetrahedral geometry. Assume that the ligands are capable of σ interactions only. How does this result for Δ_t compare with the value for Δ_o? 4
   (d) The $[Ni(CN)_4]^{2-}$ ion, which has a square-planar geometry, is diamagnetic, whereas the $[NiCl_4]^{2-}$ ion, which has a tetrahedral geometry, is paramagnetic. Show the crystal field splitting diagrams for those two complexes? 3

6. (a) Explain the terms Lability and Inertness of complexes? 4
   (b) Show the substitution reaction mechanism in $[Co(NH_3)_4CO_3]^+$ to produce $[Co(NH_3)_4(H_2O)_2]^{3+}$ without breaking the metal-ligand bond. 3
   (c) Discuss the mechanism of nucleophilic substitution reaction in square planer complexes. 3
   (d) What is trans-effect? Explain it in terms of electrostatic polarization theory. 4

7. (a) Deduce the equation $P = 1 + 4\pi\kappa$ and mention the terms in it. What are gram (χ_g) and molar (χ_m) susceptibility? 2+2=4
   (b) What are anti-ferromagnetic and ferromagnetic coupling? State Curies Law and show change of χT graphically for the above two types of interactions with variable temperature. 3+2=5
   (c) Briefly explain temperature-independent paramagnetism with examples. 2
   (d) At room temperature, the observed value of μ_{eff} for $[Cr(en)_3]Br_2$ is 4.75 μ_B. Is the complex high-or low-spin? 3

8. (a) $K_2[NiF_6]$ is diamagnetic while $K_3[CoF_6]$ is paramagnetic. Both have same d^6 configuration. 2
   (b) Explain the effect on the *d*-orbital energies when an octahedral complex is compressed along the *z* axis. 3
   (c) Explain why CFSE for a low-spin d^5 configuration contains a 2*P* term. 2
   (d) Draw a molecular orbital energy level diagram for an octahedral complex $[CoL_6]^{3-}$ showing bonding, non-bonding and anti-bonding MOs. 4
   (e) The absorption spectrum of $[Ti(H_2O)_6]^{3+}$ exhibits a band with λ_{max} = 510 nm. What colour of light is absorbed and what colour will aqueous solutions of $[[Ti(H_2O)_6]^{3+}$ appear? Why? 3

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2020
Course Title: Inorganic Chemistry III (Advanced Concepts)
Course # Chem. 320, Full Marks: 70, Time: 4 hrs.

[Answer any five questions]

1. (a) State the Zero-point energy. 2.0
(b) What is the physical interpretation of the wave function? Mention the acceptable conditions of a wave function Ψ. 4.0
(c) Briefly explain the results of the particle in one-dimensional box calculation. 5.0
(d) How many radial nodes does each of the following orbitals possess: 3.0
(i) 2s; (ii) 4s; (iii) 3p; (iv) 5d

2. (a) What is the notion of Valence bond Theory (VBT) proposed by Linus Pauling in explaining structures of coordination complexes? 2.0
(b) Draw the potential energy curve for a system containing two H atoms for the following situations: 4×1.5
(i) H atoms are at infinite distance
(ii) H atoms are approaching each other and are at a distance for electronic exchange.
Compare and explain how the potential energy curve can be improved with the following approximations:
(iii) Electronic indistinguishability (iv) Covalent-ionic contribution
(c) The orbitals are in general too large and too high in energy to take part in hybridization with s and p orbitals, and yet it happens – explain. 2.0
(d) Describe the σ-bonding in the following complexes using VBT, mention their hybridization and molecular structure. 4.0
(i) High spin $[Co(NH_3)_6]^{3+}$ (ii) $[CoCl_4]^{2-}$ (iii) $[Ni(CN)_4]^{2-}$

3. (a) What are Bonding and Anti-bonding Molecular orbitals? Write down the bonding and anti-bonding wave functions for the two MOs. What is overlap integral? 1.5+1.5+1
Draw the (a) probability function diagram for bonding Ψ_b^2 and anti-bonding Ψ_a^2.
(b) How would you account for the fact that, N_2 is unreactive and diamagnetic but molecular O_2 is paramagnetic? 2.0
(c) Provide a MO diagram for heterodiatomic molecule CO and compare bond order and bond length in CO and CO^+. 2.0
(d) (i) Define π and σ overlap. 4.0
(ii) Draw the π-overlap of a metal d orbital with the following:
(A) A ligand p orbital (B) A metal d orbital (C) ligand π* orbital. (D) ligand σ* orbital. 2.0
(e) What are ligand group orbital (LGO)?

4. (a) What are the unique properties of transition metal elements? 1.0
(b) Write down the nomenclature of the following coordination complexes: 3.0
(i) $Pt(NH_3)BrCl(CH_3NH_2)$ (ii) $[Co(NH_2CH_2CH_2NH_2)_2Cl_2]^+$ (iii) $[Co(NH_3)_4(OH)_2]^{3+}$
(c) Define denticity. When the eta (η) or kappa (κ) notations are used in nomenclature? Draw the structures of the following ligands and mention their denticity: 2+3
(i) Trien (ii) EDTA (iii) acac (iv) dppe
(d) Define isomerism. Draw structures of the *fac* and *mer* isomers of the complex $[Co(NH_3)_3Cl_3]$. Write notes on the following with examples. 2+3
(i) Linkage isomer (ii) Ionization isomer (iii) Hydrate isomer

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5. (a) Define Crystal field splitting. How would you measure it experimentally? 2.0
(b) Show the splitting of d orbitals for any divalent metal cation M^{2+} by energy level diagram for (i) Tetrahedral (ii) square planar ligand environment. 3.0
(c) Define Crystal Field Stabilization energy (CFSE). Calculate CFSE for $[V(H_2O)_6]^{3+}$ in kJ mol^{-1}. Given $\Delta_o = 18{,}900$ kJ mol^{-1}. 2+2.5
(d) What is paring energy? Calculate CFSE for high spin $[Fe(H_2O)_6]^{2+}$ complex in kJ mol^{-1}. Given $\Delta_o = 10{,}400$ cm^{-1} and P = 17,600 cm^{-1}. 1 $kJmol^{-1} = 83.7$ cm^{-1}. 2+2.5

6. (a) What do you understand by trans effect? Discuss the electronic polarization theory proposed for the trans effect. 3.0
(b) Draw the molecular orbital energy level diagram of $[Ni(NH_3)_6]^{2+}$ and mentions its magnetic nature? 3.0
(c) A metal ion M^{n+} having d^4 valence electronic configuration combined with three bidentate ligands to form a complex compound. Assuming $\Delta_o > p$. 4.0
(i) Draw the diagram showing d-orbital splitting during this complex formation,
(ii) What type of hybridization will M^{n+} have?
(iii) Name the type of isomerism exhibited by this complex.
(iv) Write the electronic configuration of metal Mn^{+}.
(d) What is John-Teller distortion? Explain, the Jahn-Teller distortion in $[Cu(H_2O)_5]^{2+}$. 4.0

7. (a) What are 'Associative' and 'Dissociative' mechanism in octahedral substitution reaction? Show the above mechanisms pictorially for an octahedral complex $[MX_6]$. 2+2
(b) What do you understand by *trans* effect? Discuss the π-bonding theory proposed For *trans effect*. 1+3
(c) Predict the products (equimolar mixtures of reactants): 4.0

(i) $[Pt(NH_3)_4]^{2+} \xrightarrow{Cl^-} A \xrightarrow{Cl^-} B$

(ii) $[PtCl_4]^{2-} \xrightarrow{NH_3} C \xrightarrow{NH_3} D$

(d) What are *inert* and *labile* complexes? 2.0

8. (a) How do you explain the blue color for Cu^{2+} (aq) solution, whereas Zn^{2+} (aq) is colorless? 2.0
(b) Cr^{2+} ion is very easily oxidized to Cr^{3+} - explain this fact in terms of crystal field theory. 2.0
(c) In the BF_3 molecule, the B-F bond is shorter than the normal B-F bond length – explain. 2.0
(d) Highly charged cations are rare – why? 2.0
(e) Tetrahedral complexes are usually of the high spin complex – explain. 2.0
(f) $[Fe(CN)_6]^{4-}$ is diamagnetic but $[FeF_6]^{3-}$ is paramagnetic – why? 2.0
(g) Anhydrous $AlCl_3$ is covalent in character but $AlCl_3.6H_2O$ is ionic – explain. 2.0

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[illegible] Part-III Examination 2019

[illegible] Chem 330, Full Marks 70, Time 4 hours

[Answer any five questions]

1. (a) Write the Schrödinger wave equation for the hydrogen atom in Cartesian coordinate system and explain the terms in it. 1
(b) Explain the significance of ψ and ψ^2. 3
(c) Explain the terms "*radial wave function*" and "*radial distribution function*". 4
(d) Use Slater's rule to estimate values of Z_{eff} for (i) a 4s (ii) a 3d electron in a V atom ($Z = 23$). 2
(e) What is "zero-point energy"? 2

2. (a) Prepare a molecular orbital energy level diagram for HCl molecule. Clearly sketch how the atomic orbitals interact to form MOs. 3
(b) Predict the magnetic properties and compare the bonding in O_2, O_2^+, O_2^- and O_2^{2-} in your discussion. 3
(c) Predict the structures of (i) XeF_2 (ii) $[XeF_5]^-$ and (iii) ClF_3 molecules using the VSEPR model. 6
(d) Is the VSEPR model consistent with linear or bent structure for $[NO_2]^-$? 2

3. (a) Draw the MO diagram of CO and NO molecules. Compare the bond order and the bond length of NO, NO^+ and NO^-. 5
(b) What is meant by 'gerade' and 'ungerade' orbitals? Clarify with examples. 2
(c) How many radial nodes do each of the following orbitals possess: (i) 3*d*; (ii) 5*d*; (iii) 6*s*; (iv) 2*p*? 2
(d) Envisage the hybridization and geometries of sp^3d^2 hybrid orbitals with examples. 3
(e) What is meant by HOMO and LUMO in chemistry? 2

4. (a) What is meant by chelate effect? Explain with an example. 3
(b) What are the two principle sets of equilibrium constants for expressing the formation of a series of complexes, ML, ML_2, ML_3 and so on? How they are related? 4
(c) Discuss the mechanisms for ligand substitution in the complexes ML_5X, where L are nonlabile ligands, X is the leaving ligand, and Y is the entering ligand. 4
(d) Discuss the two general mechanisms for the electron transfer reactions. 3

5. (a) Describe the crystal field theory for transition metal complexes. 3
(b) What is meant by crystal field splitting (Δ). Discuss the factors for which the magnitude of Δ depends on. 3
(c) Cr^{2+} ion is very easily oxidized to Cr^{3+} - why? Explain in terms of crystal field theory. 2
(d) Define high-spin and low-spin complexes. Which in each pair of the following complex is high-spin? Justify your answer: 2+4
(i) $[Cr(H_2O)_6]^{2+}$ and $[Cr(CN)_6]^{4-}$ (ii) $[Fe(H_2O)_6]^{2+}$ and $[Fe(CN)_6]^{4-}$

6. (a) Discuss Werner's Theory. What are the limitations of this theory? 3+1
(b) Define ligand. How ligands are classified in coordination chemistry. Give an example for each type. 1+3
(c) Define isomerism. Find the number and identity of all the isomers of $[Ma_2b_2c_2]$. 3
(d) What are the new formula of the following cobalt complexes? 3
(i) $CoCl_3.6NH_3$; (ii) $CoCl_3.5NH_3$ (iii) $CoCl_3.4NH_3$. Draw the structures.

7. (a) What is "Effective Atomic Number (EAN)"? Calculate the EAN for the central metal atom of the following complexes: 2+0.5×4
$[Ru(CO)_3(PPh_3)_2]$, $[V(CO)_6]$, $[Co(NH_3)_5(NO_2)Cl]Cl$, $K_3[Fe(CN)_5(NO)].2H_2O$, $[Co(NH_3)_3(NO_2)]$
(b) Draw all the steric isomers of $[Co(en)_2Cl_2]$ and linkage isomers of $[Co(NH_3)_5(NO_2)]$. 4
(c) Give the proper name for each of the following compounds: 4
(i) $[Cu(NH_3)_4]SO_4$; (ii) $Na_2[Fe(CN)_5].2H_2O$; (iii) $[Pt(acac)(NH_3)(Cl)$;
(iv) $Os(CO)_3(PPh_3)_2$.
(d) What are tripodal ligands? Give one specific example of this type of ligands. 2

8. Explain the followings (any four): 3.5×4
(a) $TiCl_4$ is colorless whereas $[Ti(H_2O)_6]^{3+}$ is violet.
(b) Anhydrous $AlCl_3$ is covalent but $AlCl_3.6H_2O$ is ionic in nature.
(c) Explain the "tetragonal distortion" and "rhombic distortion" in octahedral molecules with schematic diagrams.
(d) The lability of a complex decreases with the increase of charge on central metal ion.
(e) Electron transfer reactions in coordination complexes.
(f) How do you explain the blue color for Cu^{2+} (aq) solution, whereas Zn^{2+} (aq) is colorless?

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2018
Course Title: Inorganic Chemistry III (Advanced Concepts)
Course: Chem. 320
Full Marks: 70, Time: 4 hrs.

Answer any FIVE questions

1. (a) A particle is confined to a three-dimensional box of mass *m* and length *a* having infinitely potential walls and its total energy remains constant. Evaluate the energy values E_n of the particle and illustrate the energy levels as well as probability distribution functions of the above mentioned box.

   (b) Summarize and represent in figures: angular part of a wave function and angular probability distribution curves of a hydrogen atom.

   (c) How many radial nodes does each of the following orbitals possess: (i) 2*s*; (ii) 4*s*; (iii) 3*p*; (iv) 5*d*; (v) 1*s*; (vi)4*p*?

   (d) Comment on differences between plots of $R(r)$ against r, and $4\pi r^2 R(r)^2$ against r for each of the following atomic orditals of a H atom; (i) 1*s*; (ii) 4*s*; (iii) 3*p*.

2. (a) Draw the MO diagram of NO molecule. Compare the bond order and the bond length of NO, NO^+ and NO^-.

   (b) Describe the structure of the iodine monochloride, ICl in terms of molecular orbitals.

   (c) Draw a molecular orbital energy level diagram for HF molecule. Clearly sketch how the atomic orbitals interact to form MO.

   (d) What shapes would you expect for the species (a) IF_3 (b) NO_3^- (c)AsF_5 (d) $[AlCl_4]^-$?

   (e) Draw the bonding and antibonding orbital resulted from 2p orbitals with (i) σ interaction (ii) π interaction.

   (f) PCl_5 exists but PH_3 does not, why?

3. (a) In the valence bond theory (VBT) the wave function for a system containing two non interacting hydrogen atoms is given by $\Psi = \Psi_{A(1)}\,\Psi_{B(2)}$ where ΨA and ΨB designate wave functions of the atoms and the numbers 1 and 2 designte the two electrons. How can you improve the wave functions to account for both covalent and ionic infraction between the atoms.

   (b) Explain the of resonance in terms of VBT, why the NO bonds in NO_3 have a bond order of $1\frac{1}{3}$.

   (c) Describe the bonding in $H_2C{=}C{=}O$ in terms of valence bond theory.

4. (a) What are inner and outer orbital complexes? Give two examples each of inner and outer orbital complexes and explain the fact.

   (b) The maximum absorption of $[Ti(H_2O)_6]^{3+}$ shifts from 500 nm to 450 nm when H_2O is replaced by the ligand CN^-. Justify this shift in the expected direction.

   (c) Indicate the mode of hybridization of the central atom and the geometry of the following molecules.

   (i) $[Zn(NH_3)_4Cl_2]$ (ii) $K_2[Ni(CN)_4]$ (iii) $[Co(NH_3)_6]Cl_3$ (iv) $[Fe(CO)_5]$

2+2×2=

5. (a) Which complexes of the following pairs have a larger Δ value? Explain your answer. 4

(i) $[Fe(H_2O)_6]^{2+}$ and $[Fe(CN)_6]^{4-}$

(ii) $[Co(NH_3)_6]^{3+}$ and $[Rh(NH_3)_6]^{3+}$ 3

(b) Draw the possible isomeric structures of $[Co(NH_3)_3Cl_3]$ and $[Co(dien)_2]^{3+}$. 0.5×4=2

(c) Write down the names of the following complex or ions

(i) $Mo[(C_6H_5)_2PCH_2CH_2P(C_6H_5)_2]_2(N_2)_2$

(ii) $[Co((NH_3)_4(OH)_2)_3]^{6+}$

(iii) *mer*-$[Ru\{C_2H_5)_3P\}_3H_3]$

(iv) $[(NH_3)_5CoNH_2Co(NH_3)_4(H_2O)]Cl_5$

(d) Outline how would you apply crystal field theory to explain the five d-orbitals in an octahedral complex are not degenerate. Include in your answer an explanation of the 'barycentre'. 5

6. (a) Find out the exchange energy for a d^6 ion, both as a high-spin and as a low-spin in an octahedral complex. 2

(b) Explain why CFSE for a low-spin d^6 configuration contains a $2P$ term. 2

(c) Discuss the factors that contribute to the preference for forming either a high- or a low spin d^4 complex. 3

(d) Draw a molecular orbital energy level diagram for an octahedral complex $[CoL_6]^{3-}$ showing bonding, non-bonding and anti-bonding MOs. 4

(e) Calculate the spin only magnetic moment for a d^7 ion in tetrahedral and square planar complexes. 3

7. (a) What is Jahn- Teller distortion? In $[Cu(NH_3)_6]^{2+}$ complex four of the Cu-N distances are short and two are long, Provide an explanation. 4

(b) What is pairing energy? Calculate CFSE for high spin $[Fe(H_2O)_6]^{2+}$ complex in $KJmol^{-1}$, Given $\Delta o = 10400 KJmol^{-1}$ and $P = 17600 cm^{-1}$ 2+3=5

(c) Why is the presence of acidic proton in a ligand necessary for the fast base hydrolysis reaction of octahedral complexes? Give an account for it. 5

8. Give short answers/explanations of the following questions (any seven): 7×2=14

a) What is meant by stereochemically inert lone pair? Give an example.
b) What do you understand by denticity of ligand? Give an example.
c) What does the term 'ligand group orbital' mean?
d) Sketch the defining characteristics of π and δ molecular orbital.
e) What is meant by the chelate effect? Give an example.
f) What two factors contribute to giving high *trans* effects?
(g) Comparison of VBT and MOT
(h) Effective nuclear charge and shielding.

Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) (Part III) Final Examination 2017
Course Title: Inorganic Chemistry III (Advanced Concepts)
Course No. Chem. 320F
Time: 4 Hours, Total Marks: 70

Answer any five questions

1. (a) A particle is confined to a one-dimensional box of mass m and length a having infinitely potential walls and its total energy remains constant. Evaluate the energy values E_n of the particle and illustrate the energy levels as well as probability distribution functions of the above mentioned box. 5
(b) What is the physical interpretation of the wave function? Mention the acceptable conditions of a wave function ψ. 3
(c) Summarize and represent in figures: (i) radial part of a wave function (ii) radial probability distribution curves of the electron wave function for different electronic states of the hydrogen atoms. 4
(d) What is meant by 'gerade' and 'ungerade' orbitals? Clarify with examples. 2

2. Write the wave functions for a system containing two hydrogen atoms in the following situations using the principles of valence bond theory (VBT): 2
(a) The hydrogen atoms are not interacting. 2
(b) The hydrogen atoms start approaching each other and the electrons are exchanged to allow for covalent interaction. 2
(c) The electrons interact so that both may be simultaneously near one of the hydrogen atoms. 2
(d) Draw the potential energy curves to explain the energy changes that take place in the above situations. 2
(e) Which one of the above mentioned situations leads to the greatest stabilization of the system? 2
(f) 'The idea of delocalization of electrons and resonance have origin in valence bond theory' – Explain. 4

3. (a) The Carbonate ion CO_3^{2-} ion has a symmetrical trigonal planar structure. 3+1+1
(i) Draw appropriate resonance structures which contribute to the structure.
(ii) Find out the bond order of the C–O bonds.
(iii) Comment on the hybridization of orbitals on the carbon atom.
(b) For B_2 molecule: 2
(i) Draw the molecular orbital energy level diagram. 1
(ii) Write the molecular electronic configuration. 1
(iii) Give the bond order. 1
(iv) Comment on magnetic property. 2
(v) Draw the highest occupied molecular orbital (HOMO), and explain its symmetry. 2
(vi) Draw the lowest unoccupied molecular orbital (LUMO) and explain its symmetry.

4. (a) Which complexes of the following pairs has the larger value Δ_o? Explain your answer.
   (i) $[Cr(NH_3)_6]^{3+}$ and $[CrCl_6]^{3-}$
   (ii) $[Co(H_2O)_6]^{3+}$ and $[Rh(H_2O)_6]^{3+}$
   (iii) $[Co(H_2O)_6]^{2+}$ and $[Co(H_2O)_6]^{3+}$
   (b) Draw and name all the possible isomeric structures of $[Cr(NH_3)_3Cl_3]$ 2
   (c) What is crystal field stabilization energy? Explain the splitting of *d*-orbitals under the influence of an octahedral field. 4
   (d) Find out the exchange energy for a d^7 ion, both as a high-spin and as a low-spin in an octahedral complex. 2

5. (a) The high-spin d^4 complex $[Cr(H_2O)_6]^{2+}$ is *labile*, but the low-spin d^4 complex ion $[Cr(CN)_6]^{4-}$ is *inert*. Clarify this statement. 3
   (b) $K_2[NiF_6]$ is diamagnetic while $K_3[CoF_6]$ is paramagnetic. Both have same d^6 configuration. 3
   (c) Using the angular overlap model, determine the relative energies of d orbitals in a metal complex of formula ML_4 having tetrahedral geometry. Assume that the ligands are capable of σ interactions only. How does this result for Δ_t compare with the value for Δ_o? 4
   (d) Tetrahedral complexes are usually of high spin complex. Explain. 2
   (e) Account the observation- BI_3 is trigonal planar while PI_3 is trigonal pyramidal in shape. 2

6. (a) What do you understand by *trans effect* and *trans influence*? Do they have any differences-If have, explain with suitable examples? 7
   (b) Explain the reaction mechanism of the substitution reaction in *trans* and *cis*- $[M(LL)_2BX]$ (LL = a bidentate ligand such as en) complexes. 6
   (c) Of the first-row transition metal complexes of formula $[M(NH_3)_6]^{3+}$, which metals are predicted by the Jahn-Teller theorem to have distorted complexes. 3
   (d) What do you mean by tripodal ligands? Explain with suitable examples. 2

7. (a) Calculate the crystal field stabilization energy for a d^8 ion such as Ni^{2+} in octahedral and tetrahedral complexes. Use units of Δ_o in both cases. Which is the most stable? State any assumptions made. 4
   (b) Envisage the hybridization and geometries of sp^3d and dsp^2 hybrid orbitals with examples. 4
   (c) Write resonance structures, including formal charges, for O_3, SO_3, NO_2. 3
   (d) How will you account the fact that N_2 is unreactive but CO is highly reactive and poisonous while they are isoelectronic? 3

8. Explain any four of the following: 3.5×4
   (a) There is a mismatch of the sign of the wave function in the π-system of $[(PNCl_2)_3]$.
   (b) Cr^{2+} ion is very easily oxidized to Cr^{3+} - explain this fact in terms of crystal field theory.
   (c) The lability of a complex decreases with the increase of charge on central metal ion.
   (d) The NO stretching frequency of $[NO(HSO_4)]$ is greater than that of NO molecule.
   (e) $TiCl_4$ is colourless whereas $[Ti(H_2O)_6]^{3+}$ is violet.
   (f) The divalent monochloro cation of Cr(III), $[Co(NH_3)_5Cl]^{2+}$ aquate much slower than the monovalent dichloro cation $[Co(NH_3)_4Cl_2]^+$.
   (g) $[Fe(CN)_6]^{4-}$ is diamagnetic but $[FeF_6]^{3-}$ is paramagnetic.

DEPARTMENT OF CHEMISTRY
JAHANGIRNAGAR UNIVERSITY
B. Sc. (Part III) HONOURS EXAMINATION 2016
Course Title: *Inorganic Chemistry III (Advanced Concepts)*
Course No. Chem-320
Time: 4 Hours. Total Marks: 70

Answer any five questions

1. (a) Derive the Schrodinger equation. What is the significance of this equation? 3+1
(b) What is meant by (i) Well behaved function (ii) normalization of wave function? 3
(c) Write the Schrodinger equation for a particle in one dimensional box and that of three dimensional box.. 2
(d) Give the energy expression for a particle of mass 'm' moving in a cubic box of length 'a'. Hence (i) calculate the zero point energy; (ii) show that the 2nd energy level is degenerate. 1+4

2. (a) In valence bond theory (VBT) the wave function for a system containing two non interacting hydrogen atoms is given as $\psi A = \psi_{A(1)}\ \psi_{B(2)}$; where and ψ_A and ψ_B designate the wave functions of atoms A and B and the numbers 1 and 2 designate the two electrons. 2+2+3
(i) Write the wave function when the electrons are exchanged to allow for covalency.
(ii) Give the wave function when both the electons remain on atom A or B giving ionic structure.
(iii) Explain, by drawing the potential curves, how do the enegies change in (i) and (ii).
(b) Explain using the concept of resonance why the N-O bonds in NO_2^- have a bond order of 1½ 3
(c) Define hybridization. Why is the use of hybrid orbitals are preferable over pure atomic orbitals in forming covalent bond? 4

3. (a) What do you mean by Molecular Orbital? How does it differ from Atomic Orbital? 2
(b) Explain with suitable example the formation of MO from AO through LCAO. 2
(c) Draw the bonding and antibonding orbital resulted from 2*p* orbitals with (i) σ interaction (ii) π interaction. 2
(d) Draw the MO diagram of C_2 molecule. How does and why it differs from the MO diagram of O_2 and other molecules of elements having higher atomic number? 3
(e) Draw the MO diagram of CO molecule. Hence explain 2+1.5+1.5
(i) CO is highly poisonous (ii) It can act as π acid ligand.

4. (a) What is the strength of molecular orbital theory (MOT)? How it differs from Cyrstal field theory (CFT)? 2+1
(b) Discuss and depict the Molecualr orbital energy diagram for σ bond formation of a homonuclear diatomic molecule Li$_2$. 4
(c) What are 'Ligand Group Orbitals (LGO's)'? Draw a Molecualr orbital energy level diagram for an octahedral complex $[CoL_6]^{3+}$ showing boding, non-bonding and anti-bonding MO's. 1+4
(d) Define HOMO and LUMO. 2

5. (a) Define kinetic and thermodynamic stability. How do they correlate to activation energy E_A and ΔG^o? 2.5

(b) Draw reaction coordinate diagram for the following situations: 2.5

(i) Large activation energy with $\Delta G^o<0$.
(ii) Large activation energy with $\Delta G^o<0$ and intermediate is detectable.
(iii) Small activation energy with $\Delta G^o>0$.

(c) What are 'Associative and 'Dissociative' mechanism in octahedral substitution reactions? Show pictorially these two mechanism for an octahedral metal complex $[ML_6]$. 2+2

(d) $[AlF_6]^{3-}$, $[SiF_6]^{2-}$, $[PF_6]^-$ and SF_6 are all six coordinate complexes. Arrange them in order of decreasing ligand exchange rate and explain. 3

(e) What is the difference between an 'intermediate' and 'transition' state? 2

6. (a) Define crystal field splitting. How would you measure it experimentally?. 2

(b) Show comparative splitting of 'd' orbitals of any divalent metal cation M^{2+} by energy level diagram for (i) tetrahedral and (ii) square planar ligand environment. 3

(c) Define crystal field stabilization energy (CFSE). Calculate CFSE for $[V(H_2O)_6]^{3+}$ in kJ mol^{-1}. Given Δ_o = 18,900 kJ mol^{-1}. 2+2.5

(d) What is pairing energy? Calculate CFSE for high spin $[Fe(H_2O)_6]^{2+}$ complex in kJ mol^{-1}, given Δ_o = 10,400 kJ mol^{-1} and p = 17600 cm^{-1}. (1 kJ mol^{-1} = 83.7 cm^{-1}).

7. (a) (i) What is Jahn-Teller distortion? 2+2.5

(ii) How would you rationalize the fact that the octahedral complex $[CrF_6]^{4-}$ has four Cr-F bonds of length 1.98 – 2.01Å and two longer bonds of length 2.43 Å. 2

(b) Ni^{2+} reacts successfully with all of the following ligands CN^-, Cl^- and NH_3. One of the complexes is octahedral while the others are of square planar and tetrahedral symmetry.

(i) Arrange the ligands according to their strength. 3+3

(ii) Assign ligands to the corresponding complexes, give chemical formulae with suitable explanation.

(c) What do you understand by *trans*-effect? Predict the products (equimolar mixture of reactants): 2+2

i) $Pt(NH_3)_4^{2+}$ —Cl⁻→ A —Cl⁻→ B

ii) $PtCl_4^{2-}$ —NH_3→ C —NH_3→ D

8. (a) Define geometrical isomerism.

(b) Draw and name all the possible isomeric structures of $[Cr(NH_3)_3Cl_3]$ 1.5

(c) Why geometrical isomerism is not possible for tetrahedral complex?

(d) Give an example of each of the following:

i) a coordination of isomer of $[Co(NH_3)_6][Cr(SCN)_6]$

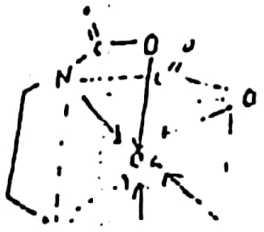
ii) an ionization isomer of $[Co(NH_3)_5Br]SO_4$

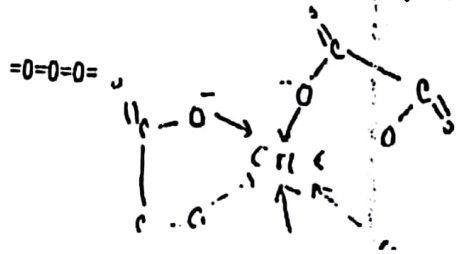
iii) A hydrate isomer of $[Co(NH_3)_4(H_2O)Cl]Br_2$

iv) A linkage isomer of $[Co(NH_3)_5(ONO)]Cl_2$

(e) What are chelating ligands? Draw the structures of the following complexes: 1.5+2

(i) Ca-EDTA; (ii) $[Cr(oxalato)_3]^{3-}$.





Department of Chemistry
Jahangirnagar University
B. Sc. (Part III) Honours Examination 2015
Course Title: *Inorganic Chemistry III (Advanced Concepts)*
Course No. Chem-320
Time: 4 Hours, Total Marks: 70

Answer any five questions

1. (a) Write the Schrodinger equation for a particle in one dimensional box and that of three dimensional box. 2.0
(b) Explain the significance of ψ and ψ^2. 3.0
(c) Briefly explain the results of particle in one dimensional box calculation. 4.0
(d) Give a brief comparison of particle in one dimensional and three dimensional box calculations. 3.0
(e) Explain the zero-point energy. 2.0

2. Following the principles of valence bond theory (VBT), write wave function for a system containing two hydrogen atoms in the following situations: 2.0
a) The hydrogen atoms are not interacting.
b) The hydrogen atoms start approaching each other and the electrons are exchanged to allow for covalent interaction. 2.0
c) The electrons interact so that both may be simultaneously near one of the hydrogen atoms giving ionic character. 2.0
d) Explain how do the energy changes in the above situation by drawing the potential energy curve. 2.0
e) Which one of the above mentioned situations leads to greatest stabilization of the system? 2.0
f) The ideas of delocalization of electrons and resonance have origin in valence bond theory (VBT) – explain. 2+2

3. (a) Discuss the molecular orbital theory for covalent bond formation of a homonuclear diatomic molecule. 4.0
(b) Draw ψ-r and ψ^2-r graph for both bonding and anti-bonding molecular orbital resulted from overlap of two 1s orbitals. 3.0
(c) Give the molecular orbital diagram of NO. Hence compare the N-O bond order and bond length of NO and NO^+. 4.0
(d) How will you account the fact that N_2 is unreactive and CO is highly reactive and poisonous-although they are isoelectronic? J.D Lee 3.0

4. (a) Write down the names of the following complex compounds or ions:
i) $[Cr(H_2O)_4Cl_2]^+$ ii) $[Co(NH_3)_5(SO_4)]^+$ iii) $K_3[Al(C_2O_4)_3]$
iv) $[Pt(NH_3)_4(NO_2)Cl]SO_4$
v) $K_4\left[(C_2O_4)_2Co\begin{matrix}OH\\OH\end{matrix}Co(C_2O_4)_2\right]$ 0.5×5=2.5

(b) Indicate the hybridization of the central metal atom; geometry and draw the structures of the following compounds. 1.5×5=7.5
i) $K_3[Fe(CN)_6]$ ii) $[Cr(en)_2]Cl_2)]$ iii) $K_2[Ni(CN)_4]$
iv) $[Zn(NH_3)_4]Cl_2$ v) $[PtCl_6]^{2-}$
Fe = 26, Cr = 24, Ni = 28, Zn = 30 and Pt = 78

6

Tutorial Examination 2017
Course: Chem. 320F, Full Marks: 20, Time: 50 Mins.

1. A particle is confined to a one-dimensional box of mass m and length a having infinitely potential walls and its total energy remains constant. Evaluate the energy values E_n of the particle and illustrate the energy levels as well as probability distribution functions of the above mentioned box. 5
2. What is the physical interpretation of the wave function? Mention the acceptable conditions of a wave function ψ. 3
3. Summarize and represent in figures: (a) radial part of a wave function (b) radial probability distribution curves of the electron wave function for different electronic states of the hydrogen atoms. 4
4. Derive the famous wave equation by Schrödinger, and how is it used or the importance? 5
5. What is meant by 'gerade' and 'ungerade' orbitals? Clarify with examples. 3

Tutorial Examination, Course: Chem 320
Marks: 20, Time: 60 Minutes, Date: 03.12.2017

1. Draw the MO diagram of CO and NO molecules. Compare the bond order and the bond length of NO, NO^+ and NO^-. 6
2. Explain the formation of MO from AO through LCAOs with suitable examples. 2
3. Predict the magnetic properties and compare the bonding in O_2, O_2^+, O_2^- and O_2^{2-} in your discussion. 3
4. Prepare a molecular orbital energy level diagram for HCl molecule. Clearly sketch how the atomic orbitals interact to form MOs. 3
5. Envisage the hybridization and geometries of sp^3d and sp^3d^2 hybrid orbitals with examples. 4
6. What is meant by HOMO and LUMO in chemistry? 2

3rd Tutorial Examination
Course: Chem. 320F, Full Marks: 20
Time: 60 Mins. Date: 13.02.2018

1. Which complexes of the following pairs have a larger Δ value? Explain your answer. 6
   (i) $[Cr(NH_3)_6]^{3+}$ and $[CrCl_6]^{3-}$.
   (ii) $[Co(H_2O)_6]^{3+}$ and $[Rh(H_2O)_6]^{3+}$
   (iii) $[Cr(H_2O)_6]^{3+}$ and $[Cr(H_2O)_6]^{2+}$
2. Draw and name all the possible isomeric structures of $[Cr(NH_3)_2Cl_3]$ 3
3. Why geometrical isomerism is not possible for tetrahedral complex? 2
4. Write down the names of the following complex or ions: 0.5×4 = 2
   (i) $Mo[(C_6H_5)_2PCH_2CH_2P(C_6H_5)_2]_2(N_2)_2$
   (ii) $[Co(NH_3)_5CO_3]Cl$
   (iii) *mer*-$[Ru\{(C_2H_5)_2P\}_3H_3]$
   (iv) $[Pt(NH_3)_3Cl][Pt(NH_3)Cl_3]$
5. What is crystal field stabilization energy? Explain the splitting of *d*-orbitals under the influence of an octahedral field. 2+3 =5
6. Find out the exchange energy for a d^3 ion, both as a high-spin and as a low-spin in an octahedral complex. 2

Kiyad Che-43

Jahangirnagar University
Department of Chemistry
3rd Year B.Sc. (Hons.)
Inorganic Chemistry-III (Advanced Concepts)
Dec 13,2022

CHEM-320F

Total marks: 20

Time: 60 min

[Answer any four of the following Questions]

1. (a) What are the postulates of Niels Bohr Quantum theory? Write down the expression for calculating E in Balmer equation. 1+1

   (b) Determine the energy of the visible emission for a transition from n_h =3 to n_l= 2 for Hydrogen atom. (R_H = 2.179 x 10^{-18} J). 03

2. (a) Write down the Schrödinger wave equation for 1- dimensional and 3-dimentional space. Write down the meaning of the terms in the equation. For what type of nuclei the Schrödinger wave equation can be solved exactly - give example. 1+1

   (b) Define ψ. Why ψ is squared? Point out the significances of ψ and ψ^2. 2+1

3. (a) What are the conditions required for a realistic solution of ψ. 3.5

   (b) Define Angular function θ(θ) ϕ(ϕ) and Radial function R(r). What part of an orbital is signified by these two factors? 1.5

4. Show that if ψ = A sin *rx* + B cos *sx* (where A, B, r and s are constants) is the wave equation for a particle in a 1-dimensional box, 5

$$\psi = A \sin rx$$

5. Plot the Radial Wave Function and the Radial Probability Functions for 2s, 2p, 3s, 3p and 3d 5 x 1.0 = 5

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Jahangirnagar University
Department of Chemistry
3rd Year B.Sc. (Hons.)

CHEM-320F
Total marks: 20

Inorganic Chemistry-III (Advanced Concepts)
Time: 50 min

[Answer any four of the following Questions]

1. (a) Draw the theoretical energy vs. internuclear distance curve for the Hydrogen molecule when two H nuclei are well separated and not interacting. Write down the overall wave function Ψ for H_A-H_B diatomic Hydrogen molecule. 03

   (b) Compare and explain how the energy curve can be improved with the following approximations (i) Electronic indistinguishability (ii) Covalent-ionic contribution. 02

2. (a) What is resonance? Give canonical structures for the CO_3^{2-} anion. Write a wave function showing contributions from all the canonical structures. 02

   (b) Define hybridization. Write down the wave functions for the two sp hybrid orbitals. Draw the electron density contour diagram for the sp^3 hybrid orbital. 03

3. (a) What are Bonding and Anti-bonding Molecular orbitals? Write down the bonding and anti-bonding wave functions for the two MOs. How they differ from each other? What is overlap integral? 2.5

   (b) Draw (a) probability function diagram for bonding Ψ_b^2 and anti-bonding Ψ_a^2. (b) MO Energy level diagram for H_2 molecule. 2.5

4. (a) Draw the following orbital overlap
   II) d-p and d-d whence s>0 II) d-d whence s<0 and III) p-p and d-s whence s = 0 2.5

   (b) Draw the MO energy diagram for Dioxygen and rationalize its singlet nature. 2.5

5. (a) What are frontier Molecular orbitals (FMO)? Draw the MOED B_2 and show that molecular Boron is paramagnetic. 03

   (b) Rationalize the fact that NO molecule has lower bond order than its oxidized species NO^+ 02

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Department of Chemistry (Course# 320)
3rd tutorial exam- Dated at 16-01-2020
20 marks (1 hr)

1. Define hydration energy? Why transition elements have high hydration energy? (1 +2)
2. Which of the following ions is more stable? Justify your answer (1+2)

   a) Cu^{2+} or b) Cu^{+}

3. How do you explain the blue color for Cu^{2+} (aq) solution, whereas Zn^{2+} (aq) is colorless? 2
4. Which of the following ions is more difficult to oxidize-Why? 2

   a) Mn^{2+} or b) Fe^{2+}

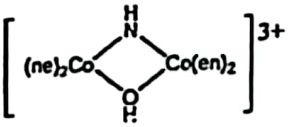
5. Write down the electron configuration for Nickel. Calculate the shielding constant for 3d and 4s electron. On the basis of shielding constant, explain, which electron (3d or 4s) should remove first for the first ionization? 5

6. Define the following terms- a) Ionization energy b) Electron affinity c) Pseudo Transition metals. (2+2+1)

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Tutorial Examination 2015
Course: Chem.320
Total Marks: 20. Time: 40 Mins.

1. Write down the names of the following coordination compounds. 1×5=5
   (i) $[Fe(H_20)_5(NO)]^{2+}$ (ii) $Na_2[Fe(CO)_4]$ (iii) $Na[Co(CO)_4]$
   (iv) $[Co(NH_3)_5Cl]Cl_2$ (v) $K_3[Al(C_2O_4)_3]$

   (vi) 

2. Define primary valency and secondary valency. Draw the structure of the complex $CoCl_3.3NH_3$ giving primary and secondary valency. 1+1+2=4
3. Define effective atomic number. 2+3=5
   Calculate the effective atomic number for the following compounds
   (i) $[Co(NH_3)_6]^{3+}$ (z=27) (ii) $[Fe(CN)_6]^{3-}$ (z=26) (iii) $[PdCl_4]^{2-}$(z=46)
4. Write down the isomer for the complexes 2
   (i) $[Co(NH_3)_5Br]SO_4$ (ii) $[Co(NH_3)_4(H_2O)Cl]Cl_2$
5. Predict and draw the shape and the hybridization for the following complexes. 4
   (i) $[Zn(NH_3)_4]^{2+}$ (z=30) (ii) $[Fe(CN)_6]^{3-}$ (Fe=26) (iii) $[PtCl_6]^{2-}$ (iv) $[Fe(F_6)]^{3-}$
   (v) [Ni (Co)4)]

=0=0=0=

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3rd Tutorial Examination
Course: Chem. 320F, Full Marks: 20
Time: 60 Mins. Date: 04.02.2019

1. Which complexes of the following pairs have a larger Δ value? Explain your answer. 4
   (i) $[Fe(H_2O)_6]^{2+}$ and $[Fe(CN)_6]^{4-}$
   (ii) $[Co(NH_3)_6]^{3+}$ and $[Rh(NH_3)_6]^{3+}$
2. Draw the possible isomeric structures of $[Co(NH_3)_3Cl_3]$ and $[Co(dien)_2]^{3+}$. 3
3. Explain why CFSE for a high-spin d^5 configuration contains a $2P$ term. 2
4. Write down the names of the following complex or ions 0.5×4 = 2
   (i) $Mo[(C_6H_5)_2PCH_2CH_2P(C_6H_5)_2]_2(N_2)_2$
   (ii) $[Co(Co(NH_3)_4(OH)_2)_3]^{6+}$
   (iii) *mer*-$[Ru(C_2H_5)_3P]_3H_3$
   (iv) $[(NH_3)_5CoNH_2Co(NH_3)_4(H_2O)]Cl_5$
5. Outline how you would apply crystal field theory to explain the five *d*-orbitals in an octahedral complex are not degenerate. Include in your answer an explanation of the 'barycentre. 5
6. Find out the exchange energy for a d^5 ion, both as a high-spin and as a low-spin in an octahedral complex. 2
7. The absorption spectrum of $[Ti(H_2O)_6]^{3+}$ exhibits a band with λ_{max} = 510 nm. What colour of light is absorbed and what colour will aqueous solutions of $[Ti(H_2O)_6]^{3+}$ appear? Why? 2

yellow gelb green.

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MO Orbital theory Orgula

2nd Tutorial Examination
Course: Chem. 320F, Full Marks: 20
Time: 60 Mins. Date: 17.12.2018

1. Draw the MO diagram of NO molecule. Compare the bond order and the bond length of NO, NO^+ and NO^-. 3.5
2. Describe the structure of the iodine monochloride, ICl in terms of molecular orbitals. 2.5
3. Draw a molecular orbital energy level diagram for HF molecule. Clearly sketch how the atomic orbitals interact to form MO. 3
4. Predict the structures of (a) XeF_2, (b) $[XeF_5]^-$ and (c) SF_4 molecules using the VSEPR model. 7.5
5. What shapes would you expect for the species (a) IF_5 (b) NO_3^- (c) AsF_5 (d) $[AlCl_4]^-$? 2
6. PCl_5 exists but PH_5 does not, why? 1.5

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1st Tutorial Examination

Course: Chem 320[illegible], Full Marks: 20

Time: 50 Mins. Date: 21.07.2019

(a) A particle is confined to a one-dimensional box of mass m and length a having infinitely potential walls and its total energy remains constant. Evaluate the energy values E_n of the particle and illustrate the energy levels as well as probability distribution functions of the above mentioned box. 5

(b) What is the physical interpretation of the wave function? Mention the acceptable conditions of a wave function ψ. 3

(c) What is meant by 'gerade' and 'ungerade' orbitals? Clarify with examples. 2

(d) Use Slater's rule to estimate values of Z_{eff} for (i) a $4s$ and (b) a $3d$ electron in a V atom (Z =23). 3

(e) Interpreting radial distribution functions for $2s$ and $2p$ wave functions. Which orbital gives the electron a greater probability of close approach to the nucleus? 2

(f) How many radial nodes does each of the following orbitals possess: (i) $3d$; (ii) $5d$; (iii) $6s$; (iv) $2p$? 2

(g) Derive the famous wave equation by Schrödinger, and how is it used or the importance? 3

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Department of Chemistry (Course# 320)

4th tutorial exam- Dated at 11-02-2020

20 marks (1 hr)

1. What is meant by the chelate effect? Explain with example. 3
2. For what sizes is the chelate effect most important? How do you explain it? 1+2
3. What is meant by the term trans effect? Explain. 4
4. Discuss the two general mechanisms for electron-transfer reactions with example. 4
5. Explain any two of the following (a) Substitution reaction in Octahedral Complexes (b) Substitution reaction in Square Complexes (c) Usefulness of trans effect for the synthesis of cis and trans isomers. 3+3

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Tutorial Examination 2015
Course: Chem. 320
Total Marks: 20, Time: 40 Mins.

1. Answer the following questions on the basis of VBT:
   (a) Write the wave function when the electrons are exchanged to allow for covalence. 2
   (b) Give the wave function when both the electrons remain on either of the atoms giving ionic structure. 2
   (c) The carbonate ion CO_3^{2-} has a symmetrical trigonal planar structure. 3+0.5+0.5
      (i) Draw the resonance structure which contribute to this structure.
      (ii) Find out the bond order of C–O bonds.
      (iii) Comment on the hybridization of orbitals on the carbon atom.
   (d) Helium molecule does not exist as a diatomic molecule – explain with the help of VBT. 2

2. (a) Briefly explain the covalent bond formation in a homonuclear diatomic molecule. 4
   (b) Use molecular orbital theory to predict the bond order as well as magnetic properties of the following species: 2
   O_2^{2-}, O_2^-, O_2, O_2^+
   (c) For the B_2 molecule 2+1+0.5+0.5
      (i) Draw the molecular orbital energy level diagram.
      (ii) Write the molecular electronic configuration.
      (iii) Give the bond order.
      (iv) Comment on the magnetic property.

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Tutorial Examination 2015
Course: Chem. 320
Total Marks: 20, Time: 40 Mins.

320

1. Write down the important features of crystal field theory. 4
2. Describe the splitting of d-orbitals under the influence of octahedral field. 6
3. Explain crystal field stabilization energy. 2
4. Explain inner orbital and outer orbital octahedral complexes with examples. 2
5. Give the number of unpaired electrons(n) in the following complex ions and their magnetic behavior : 3×2
   $[Fe(CN)_6]^{4-}$ and $[Fe(CN)_6]^{3-}$

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Tutorial Examination 2017
Course: Chem. 320F, Full Marks: 20, Time: 50 Mins.

1. A particle is confined to a one-dimensional box of mass m and length a having infinitely potential walls and its total energy remains constant. Evaluate the energy values E_n of the particle and illustrate the energy levels as well as probability distribution functions of the above mentioned box. 5
2. What is the physical interpretation of the wave function? Mention the acceptable conditions of a wave function ψ. 3
3. Summarize and represent in figures: (a) radial part of a wave function (b) radial probability distribution curves of the electron wave function for different electronic states of the hydrogen atoms. 4
4. Derive the famous wave equation by Schrödinger, and how is it used or the importance? 3
5. What is meant by 'gerade' and 'ungerade' orbitals? Clarify with examples. 3

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Tutorial Examination. Course: Chem 320
Marks: 20. Time: 60 Minutes. Date: 03.12.2017

1. Draw the MO diagram of CO and NO molecules. Compare the bond order and the bond length of NO, NO^+ and NO^-. 6
2. Explain the formation of MO from AO through LCAOs with suitable examples. 3
3. Predict the magnetic properties and compare the bonding in O_2, O_2^+, O_2^- and O_2^{2-} in your discussion.
4. Prepare a molecular orbital energy level diagram for HCl molecule. Clearly sketch how the atomic orbitals interact to form MOs.
5. Envisage the hybridization and geometries of sp^3d and sp^3d^2 hybrid orbitals with examples. 4
6. What is meant by HOMO and LUMO in chemistry? 2

11

1st Tutorial Examination
Course: Chem 320F, Full Marks: 20
Time: 50 Mins. Date: 31.10.2018

1. (a) A particle is confined to a three-dimensional box of mass m and length a having infinite potential walls and its total energy remains constant. Evaluate the energy values of the particle and illustrate the energy levels as well as probability distribution functions of the above mentioned box. 5

(b) Summarize and represent in figures: angular part of a wave function and angular probability distribution curves of a hydrogen atom. 4

(c) How many radial nodes does each of the following orbitals possess: (i) 2s; (ii) 4s; (iii) 3p; (iv) 5d; (v) 1s; (vi) 4p? 3

(d) Comment on differences between plots of $R(r)$ against r, and $4\pi r^2 R(r)^2$ against r for each of the following atomic orbitals of an H atom: (i) 1s; (ii) 4s; (iii) 3p. 3

(e) What is the physical interpretation of the wave function? Mention the acceptable conditions of a wave function ψ. 3

(f) What is meant by 'gerade' and 'ungerade' orbitals? Clarify with examples. 2

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2nd Tutorial Examination
Course: Chem. 320F, Full Marks: 20
Time: 50 Mins. Date: 02.09.2019

1. Draw the MO diagram of NO molecule. Compare the bond order and the bond length of NO, NO^+ and NO^-. 3
2. Predict the magnetic properties and compare the bonding in O_2, O_2^+, O_2^- and O_2^{2-} in your discussion. 3
3. Prepare a molecular orbital energy level diagram for HCl molecule. Clearly sketch how the atomic orbitals interact to form MOs. 3
4. Predict the structures of (a) XeF_2, (b) $[XeF_5]^-$ and (c) ClF_3 molecules using the VSEPR model. 6
5. Is the VSEPR model consistent with a linear or bent structure for $[NO_2]^+$? 2
6. Envisage the hybridization and geometries of sp^3d^2 hybrid orbitals with examples. 3

7. Find out the exchange energy for a d^4 ion, both as a high-spin and as a low-spin in an octahedral complex. 2
8. The absorption spectrum of $[Ti(H_2O)_6]^{3+}$ exhibits a band with $\lambda_{max} = 510$ nm. What colour of light is absorbed and what colour will aqueous solutions of $[Ti(H_2O)_6]^{3+}$ appear? Why? 2

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Jahangirnagar University
Department of Chemistry
3rd Year B.Sc. (Hons.)

CHEM-320F — Inorganic Chemistry-III (Advanced Concepts)
Total marks: 20 — Time: 50 min

[Answer any four of the Questions]

1. (a) What are the unique properties of transition metal elements? 02
   (b) Write down the nomenclature of the following coordination complexes: 03
   (2) $Pt(NH_3)BrCl(CH_3NH_2)$ (2) $[Co(NH_2CH_2CH_2NH_2)_2Cl_2]^+$ (3) $[Co(Co(NH_3)_4(OH)_2)_3]^{6-}$

2. (a) What are the postulates of Werner's theory? Write a few limitations of Werner's theory. 2−1=3
   (b) Draw the structures of the following ligands and mention the denticity: 2
   (2) Trien (2) EDTA (3) acac (4) dppe

3. (a) Define isomerism. Draw structures of the *fac* and *mer* isomers of the complex $[Co(NH_3)_3Cl_3]$ 1.5
   (b) Define with examples: 3.5
   (1) Linkage isomer (2) Ionization isomer (3) Hydrate isomer

4. (a) The value of stability constant for an ammonia complex of Ni(II) is 7.98 whereas with ethylenediamine (en) is 18.2- explain. 2.5
   (b) The stability of $[Ni(NH_3)_6]^{2+}$ is much higher than $[Ni(H_2O)_6]^{2+}$ - explain. 2.5

5. (a) Define Hepticity and Denticity. When κ or η notations are used in naming of complexes. Elaborate with examples. 2
   (b) Write three examples of medicinal applications of coordination complexes with real examples. 3

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(1) Briefly discuss the factors that affect the crystal field splitting energy (Δ). 3
(2) Which complexes of the following pairs has a larger Δ value? Explain your answer. 6

(i) $[Cr(NH_3)_6]^{3+}$ and $[CrCl_6]^{3-}$

(ii) $[Co(H_2O)_6]^{3+}$ and $[Rh(H_2O)_6]^{3+}$

(iii) $[Cr(H_2O)_6]^{3+}$ and $[Cr(H_2O)_6]^{2+}$

(3) Explain the formation of the following on the basis of CFT: 6
(i) paramagnetic $[FeF_6]^{3-}$, (ii) diamagnetic $[Ni(CN)_4]^{2-}$, (iii) tetrahedral $[NiCl_4]^{2-}$
(4) What are the high spin and low spin complexes? 2
(5) Mode of splitting of d-orbitals in an octahedral field is just reverse of that in tetrahedral field – explain. 3

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Chem 320 Tutorial Examination: Coordination Chemistry
Full Marks 20. Date: 1-03-2015

1. Draw the structure of each of the following coordination compounds or ions. Then draw the structure for each geometrical or optical isomer that is possible. 4.5

i. *cis*-Dichlorotetraaquachromium(III) chloride
ii. *mer*-Trihydridotris(triphenylphosphine)ruthenium(III)
iii. Potassium trioxalatocobaltate(III)

2. The complex $[Co(NH_3)_6]^{3+}$ is diamagnetic but $[CoF_6]^{3-}$ is paramagnetic with four unpaired electrons. Explain these experimental observations using valence bond theory. 2.5

3. Determine the configuration(in the form $t_{2g}^{p}e_g^{q}$, as appropriate), the number of unpaired electrons, and the ligand field stabilization energy as a multiple of Δ_o or Δ_t for each of the following complexes using the spectrochemical series to decide, where relevant, which are likely to be strong-field and which weak-field. 10

(a) $[Co(NH_3)_6]^{3+}$
(b) $[Ni(OH_2)_6]^{2+}$
(c) $[Fe(CN)_6]^{3-}$
(d) $[CrF_6]^{3-}$
and
(f) $[Ni(CO)_4]$

4. Which of the following complexes would you expect to show a Jahn-Teller distortion: 3

$[CrF_6]^{4-}$ and $[Cr(CN)_6]^{4-}$.
Give reasons for your answer.

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Department of Chemistry (Course# 320)

4th tutorial exam- Dated at 11-02-2020

20 marks (1 hr)

1. What is meant by the chelate effect? Explain with example. 3
2. For what sizes is the chelate effect most important? How do you explain it? 1+2
3. What is meant by the term trans effect? Explain. 4
4. Discuss the two general mechanisms for electron-transfer reactions with example. 4
5. Explain any two of the following (a) Substitution reaction in Octahedral Complexes 3+3
   (b) Substitution reaction in Square Complexes (c) Usefulness of trans effect for the synthesis of cis and trans isomers.

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Jahangirnagar University
Department of Chemistry
3rd Year B.Sc. (Hons.)
Inorganic Chemistry-III (Advanced Concepts)

CHEM-320F
Total marks: 20

Time: 50 min

[Answer all the Questions]

1. (a) What are the unique properties of transition metal elements? 02

   (b) Write down the nomenclature of the following coordination complexes: 03
   (1) $Pt(NH_3)BrCl(CH_3NH_2)$ (2) $[Co(NH_2CH_2CH_2NH_2)_2Cl_2]^+$ (3) $[Co(Co(NH_3)_4(OH)_2)_3]^{6+}$

2. (a) What are the postulates of Werner's theory? 3

   (b) Draw the structures of the following ligands and mention the denticity: 2
   (1) Trien (2) EDTA (3) acac (4) dppe

3. (a) Define isomerism. Draw structures of the *fac* and *mer* isomers of the complex $[Co(NH_3)_3Cl_3]$ 1.5

   (b) Define with examples: 3.5
   (1) Linkage isomer (2) Ionization isomer (3) Hydrate isomer

4. (a) The value of stability constant for an ammonia complex of Ni(II) is 7.98 whereas with ethylenediamine (en) is 18.2- explain. 2.5

   (b) The stability of $[Ni(NH_3)_6]^{2+}$ is much higher than $[Ni(H_2O)_6]^{2+}$ - explain. 2.5

Jahangirnagar University
Department of Chemistry
3rd Year B.Sc. (Hons.)
Inorganic Chemistry-III (Advanced Concepts)

CHEM-320F
Total marks: 20

Time: 50 min

[Answer all the Questions]

1. (a) What is wave function? Write down the wave function Ψ for H_A-H_B diatomic Hydrogen molecule. 02

   (b) Draw the theoretical energy curve for the Hydrogen molecule. Compare and explain how the energy curve improved with the following approximations 03
   (i) Electronic indistinguishability (ii) Covalent-ionic contribution.

2. (a) What is resonance? Give canonical structures for the CO_3^{2-} anion. Write a wave function showing contributions from all the canonical structures. 3

   (b) Define hybridization. Draw the electron density contour diagram for the sp^3 hybrid orbital. 2

3. (a) What are Bonding and Anti-bonding Molecular orbitals? Write down the bonding and anti-bonding wave functions for the two MOs. What is overlap integral? 2.5

   (b) Draw the (a) probability function diagram for bonding Ψ_b^2 and anti-bonding Ψ_a^2. (b) Energy level MO diagram for H_2 molecule. 2.5

4. (a) Why the wave function Ψ is squared? Point out the significance of Ψ and Ψ^2. 1+2

   (b) Draw the energy level diagram for Dioxygen 02

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1st Tutorial Examination
Course: Chem 320P, Full Marks: 20
Time: 50 Mins. Date: 31.10.2018

1. (a) A particle is confined to a three-dimensional box of mass m and length a having infinitely potential walls and its total energy remains constant. Evaluate the energy values E_n of the particle and illustrate the energy levels as well as probability distribution functions of the above mentioned box. 5

(b) Summarize and represent in figures: angular part of a wave function and angular probability distribution curves of a hydrogen atom. 4

(c) How many radial nodes does each of the following orbitals possess: (i) $2s$; (ii) $4s$; (iii) $3p$; (iv) $5d$; (v) $1s$; (vi) $4p$? 3

(d) Comment on differences between plots of $R(r)$ against r, and $4\pi r^2 R(r)^2$ against r for each of the following atomic orbitals of an H atom: (i) $1s$; (ii) $4s$; (iii) $3p$. 3

(e) What is the physical interpretation of the wave function? Mention the acceptable conditions of a wave function ψ. 3

(f) What is meant by 'gerade' and 'ungerade' orbitals? Clarify with examples. 2

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